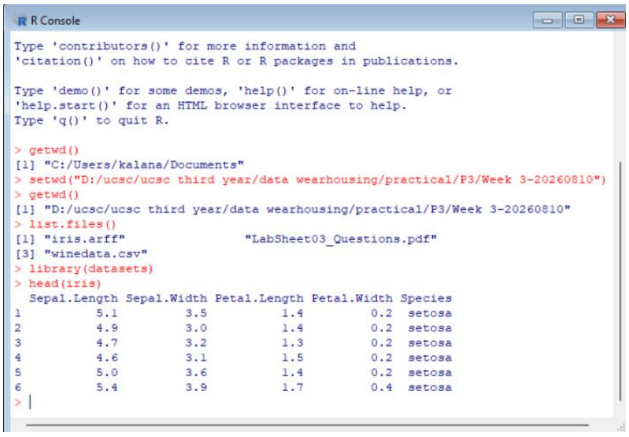
Lab Report 3

Index-23000422

Part A — K-Means on the Iris Dataset

Question 1. Loading and inspecting the data

(a) Load the built-in datasets library and display the first six rows of the iris dataset.



```
R Console

Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> getwd()
[1] "C:/Users/kalana/Documents"
> setwd("D:/ucsc/ucsc third year/data wearhousing/practical/P3/Week 3-20260810")
> getwd()
[1] "D:/ucsc/ucsc third year/data wearhousing/practical/P3/Week 3-20260810"
> list.files()
[1] "iris.arff"                "LabSheet03_Questions.pdf"
[3] "winedata.csv"
> library(datasets)
> head(iris)
  Sepal.Length Sepal.Width Petal.Length Petal.Width Species
1          5.1         3.5          1.4         0.2  setosa
2          4.9         3.0          1.4         0.2  setosa
3          4.7         3.2          1.3         0.2  setosa
4          4.6         3.1          1.5         0.2  setosa
5          5.0         3.6          1.4         0.2  setosa
6          5.4         3.9          1.7         0.4  setosa
>
```

(b) List the column (variable) names of iris. Which column is the class label, and how many classes does it have?

```
> names(iris)
[1] "Sepal.Length" "Sepal.Width"  "Petal.Length" "Petal.Width"  "Species"
>
```

class label Colum is species

```
> unique(iris$Species)
[1] setosa     versicolor virginica
Levels: setosa versicolor virginica
> length(unique(iris$Species))
[1] 3
>
```

how many classes does it have

(c) Create two new variables: x containing only the four numeric feature columns, and y containing the Species column

```
> x <- iris[, 1:4]
> y <- iris$Species
> head(x)
  Sepal.Length Sepal.Width Petal.Length Petal.Width
1          5.1         3.5          1.4         0.2
2          4.9         3.0          1.4         0.2
3          4.7         3.2          1.3         0.2
4          4.6         3.1          1.5         0.2
5          5.0         3.6          1.4         0.2
6          5.4         3.9          1.7         0.4
> head(y)
[1] setosa setosa setosa setosa setosa setosa
Levels: setosa versicolor virginica
>
```

Question 2. Running your first k-means model

a) Fit a k-means model to x using 3 clusters. Store the result in a variable called kc.

```
> set.seed(123)
>
> kc <- kmeans(x, centers = 3)
>
```

The set.seed(123) makes your result reproducible. If you run the same code again, you should get the same result.

b) Print kc and identify:

(i) the size of each cluster

Cluster 1 = 50
Cluster 2 = 62
Cluster 3 = 38

(ii)the cluster means

```
> kc
K-means clustering with 3 clusters of sizes 50, 62, 38

Cluster means:
  Sepal.Length Sepal.Width Petal.Length Petal.Width
1     5.006000    3.428000     1.462000    0.246000
2     5.901613    2.748387     4.393548    1.433871
3     6.850000    3.073684     5.742105    2.071053
```

So this is the centroids for each cluster colums

For example, if Cluster 1's centroid is:

5.006 3.428 1.462 0.246

that means the **average flower in Cluster 1** has approximately:

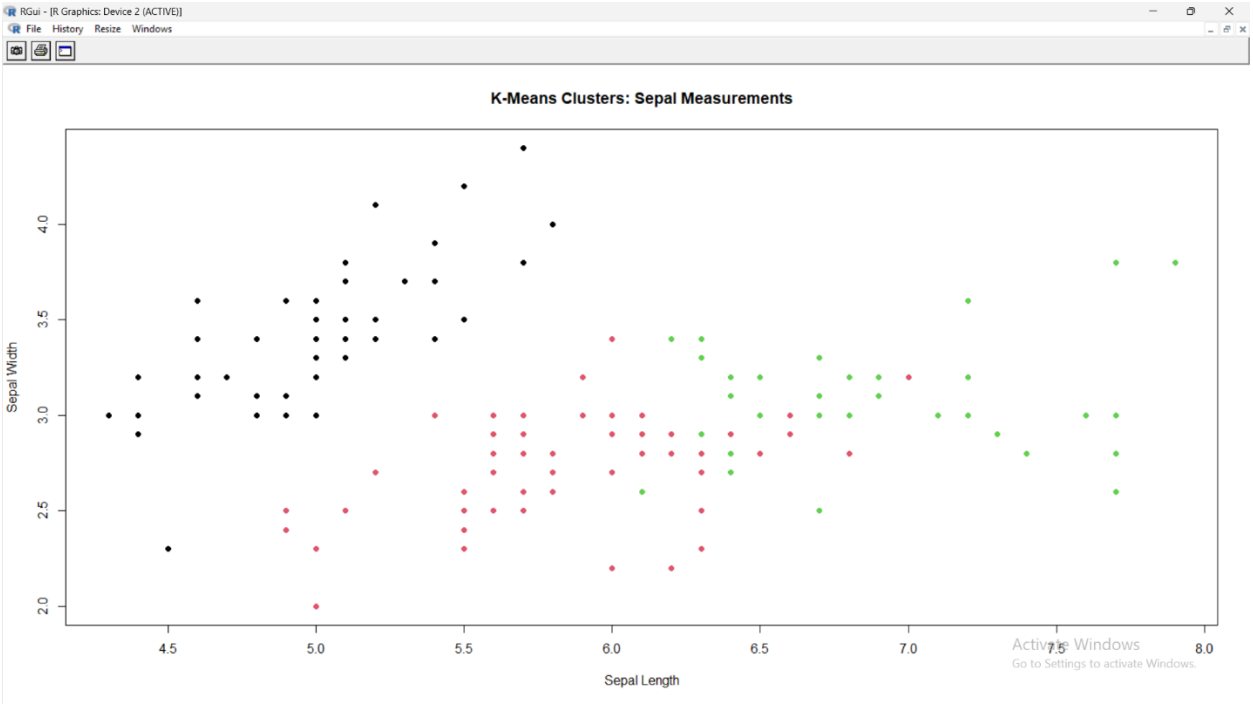
- Sepal Length = 5.006
- Sepal Width = 3.428
- Petal Length = 1.462
- Petal Width = 0.246

The cluster centres show clear differences in the measurements, particularly in petal length and petal width. Cluster 1 has relatively small petal measurements, while Cluster 3 has the largest petal length and petal width. This indicates that the petal characteristics contribute strongly to the separation of the iris observations into clusters.

(iii) the between_SS / total_SS percentage.

For each cluster, K-Means has a **center (centroid), These numbers measure how spread out each cluster is**

```
Clustering vector:
  [1] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 3 2 2 2
```



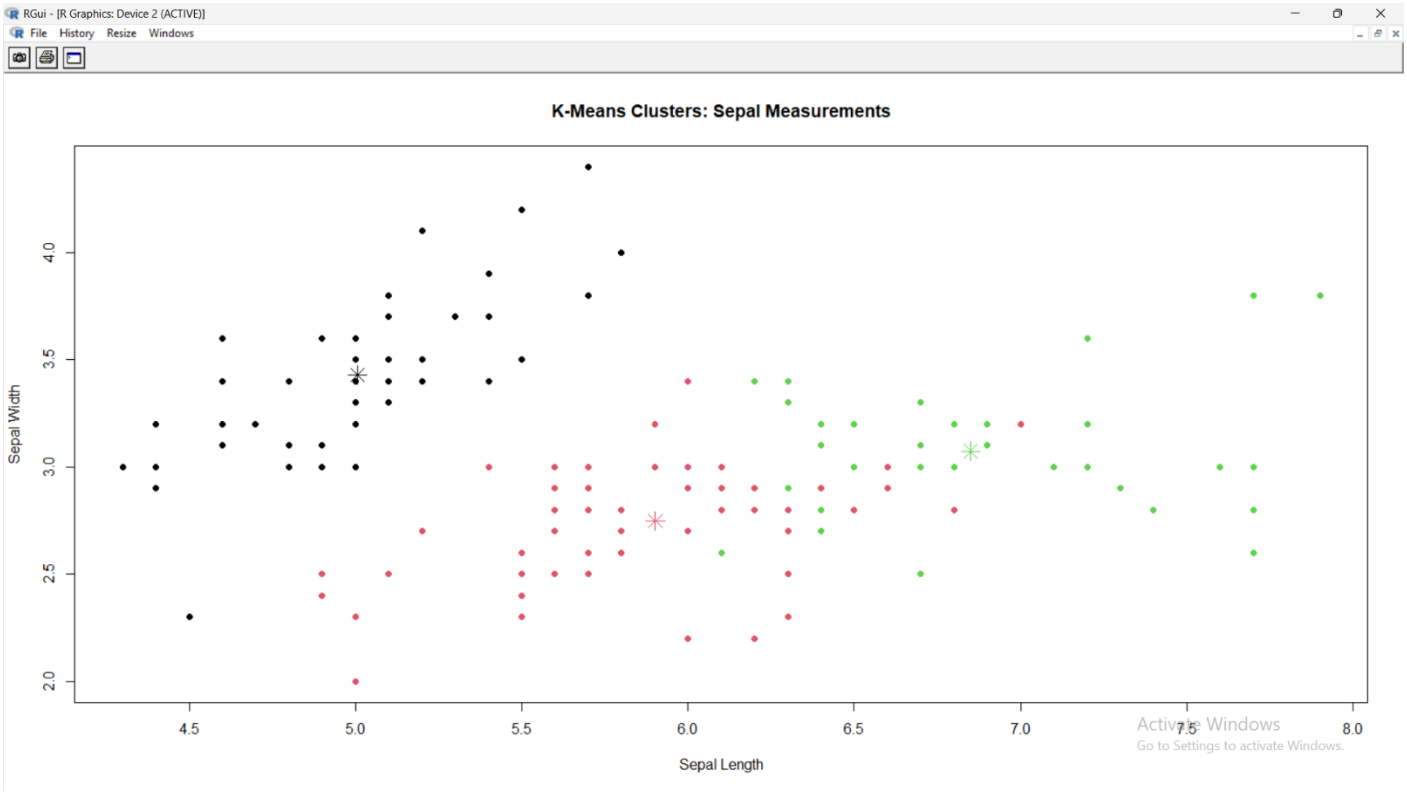
(b) Overlay the cluster centres on the same plot using points() with a distinct plotting symbol.

So stars represent the 3 centroid values coordinates

As example:

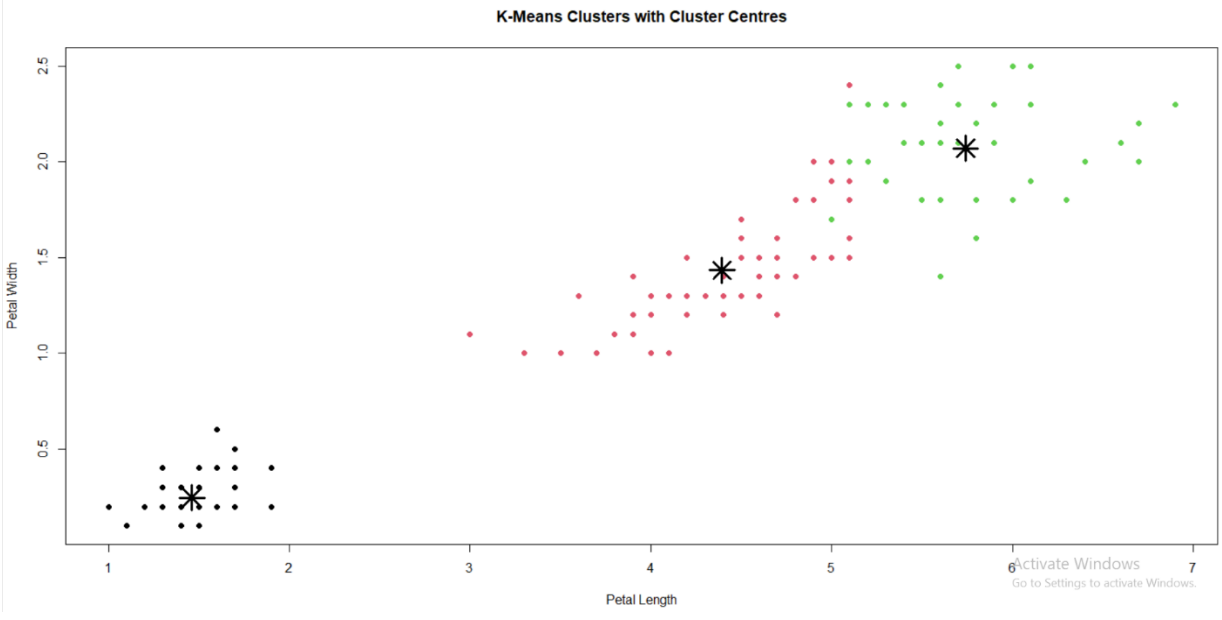
Cluster 1 have sepal.length:5.006 and sepal.width: 3.428, so these are the centroids, so that two value are one coordinates, that display by star

```
> points(
+   kc$centers[, 1],
+   kc$centers[, 2],
+   pch = 8,
+   cex = 2,
+   col = 1:3
+ )
>
```



(b) Repeat the plot using Petal.Length and Petal.Width instead. Which pair of variables gives visually cleaner clusters? Explain briefly why this makes biological sense.

```
> plot(
+   x$Petal.Length,
+   x$Petal.Width,
+   col = kc$cluster,
+   pch = 19,
+   xlab = "Petal Length",
+   ylab = "Petal Width",
+   main = "K-Means Clusters with Cluster Centres"
+ )
> points(
+   kc$centers[, 3],
+   kc$centers[, 4],
+   pch = 8,
+   cex = 3,
+   lwd = 3
+ )
>
```



Petal.Length and Petal.Width give much cleaner cluster separation .

The Petal.Length and Petal.Width pair gives visually cleaner clusters than Sepal.Length and Sepal.Width. This makes biological sense because petal size differs more substantially among the three Iris species, making petal length and width more useful features for distinguishing the species.

Question 4. Reproducibility and initial centroids

a) Explain, in your own words, why running kmeans() twice on the same data can give two different results.

Running kmeans() twice on the same data can produce different results because the **algorithm starts with randomly selected initial cluster centres**. Different starting centres can lead to different assignments of observations to clusters and therefore different final cluster centres and within cluster sum of squares. Setting a random seed using set.seed() makes the results reproducible.

b) Set a random seed of your choice with set.seed(), then re-run k-means on iris[,3:4] with 3 clusters and nstart = 20. What does the nstart argument do, and why is it good practice to use it?

```
> set.seed(123)
>
> kc2 <- kmeans(iris[, 3:4], centers = 3, nstart = 20)
> kc2
K-means clustering with 3 clusters of sizes 52, 48, 50

Cluster means:
  Petal.Length Petal.Width
1     4.269231    1.342308
2     5.595833    2.037500
3     1.462000    0.246000

Clustering vector:
  [1] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
 [38] 3 3 3 3 3 3 3 3 3 3 3 3 3 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
 [75] 1 1 1 2 1 1 1 1 1 2 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 1 2 2 2 2 2
[112] 2 2 2 2 2 2 2 2 1 2 2 2 2 2 2 1 2 2 2 2 2 2 2 2 2 2 1 2 2 2 2 2 2 2 2 2
[149] 2 2

Within cluster sum of squares by cluster:
[1] 13.05769 16.29167  2.02200
 (between_SS / total_SS =  94.3 %)

Available components:

[1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss"
[6] "betweenss"    "size"         "iter"         "ifault"
> |
```

What does nstart = 20 mean?

Instead of running K-means from **one random starting position**, R runs it **20 times with different initial cluster centres**.

It then chooses the solution with the **lowest total within-cluster sum of squares**.

c) Compare the resulting clustering to the true Species labels using table(). How many observations are misclassified?

```
> table(iris$Species, kc2$cluster)

              1  2  3
  setosa      0  0 50
  versicolor 48  2  0
  virginica   4 46  0
> 
```

Step 1 — Identify the main cluster for each species

Because K-means cluster numbers are arbitrary, we match each species with the cluster containing most of its observations

Setosa: 0 + 0 + 50

→ Cluster 3

So: 0 misclassified

Versicolor: 48 + 2 + 0

→ Cluster 1 is the main cluster.

So: 2 misclassified

Virginica: 4 + 46 + 0

→ Cluster 2 is the main cluster.

So: 4 misclassified

Total misclassified :6

correctly classified: 144

(144/150)*100=96%

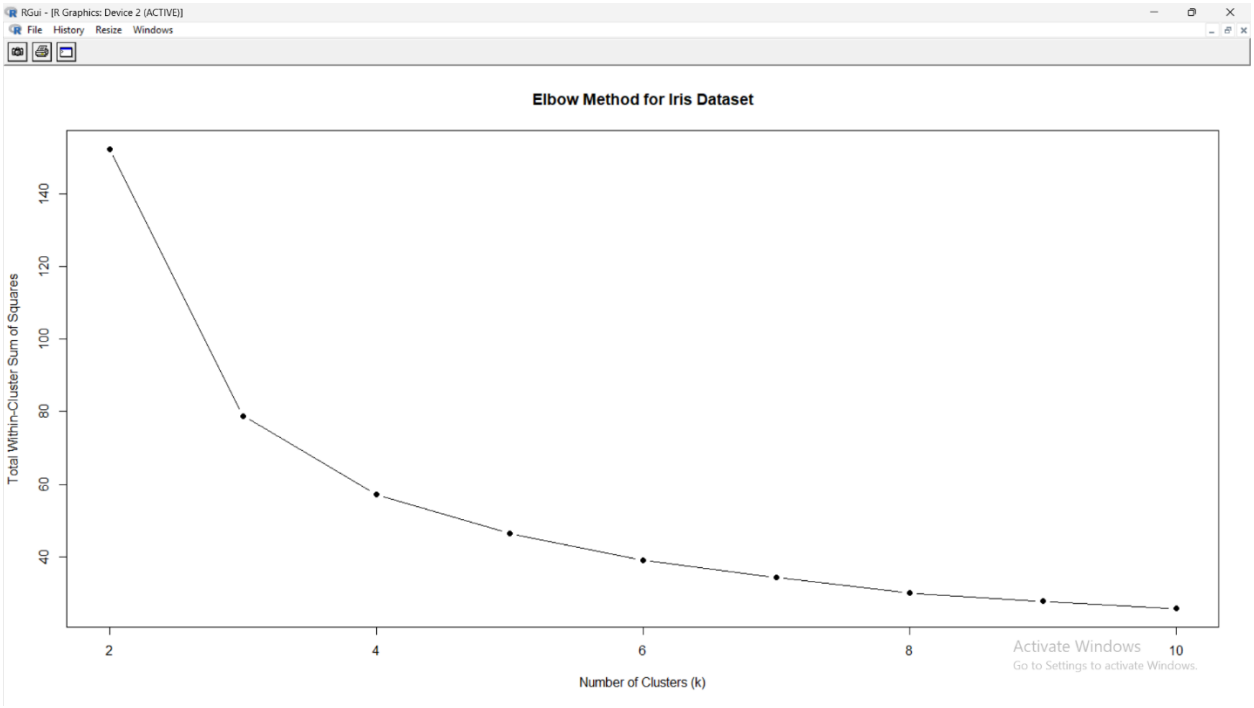
giving an effective clustering accuracy of 96%.

generally, fewer misclassified observations is better when you are comparing a clustering result against known class labels

Question 5. Choosing the number of clusters

a) Using the elbow method, compute the total within-cluster sum of squares (tot.withinss) for k = 2 to k = 10 on the iris feature data, and plot k against the within-cluster sum of squares.

```
> wss <- numeric(9)
>
> for (k in 2:10) {
+   set.seed(123)
+   km <- kmeans(x, centers = k, nstart = 20)
+   wss[k - 1] <- km$tot.withinss
+ }
> wss
[1] 152.34795  78.85144  57.22847  46.44618  39.03999  34.29823  29.98894  27.78609  25.83405
> plot(
+   2:10,
+   wss,
+   type = "b",
+   pch = 19,
+   xlab = "Number of Clusters (k)",
+   ylab = "Total Within-Cluster Sum of Squares",
+   main = "Elbow Method for Iris Dataset"
+ )
>
```



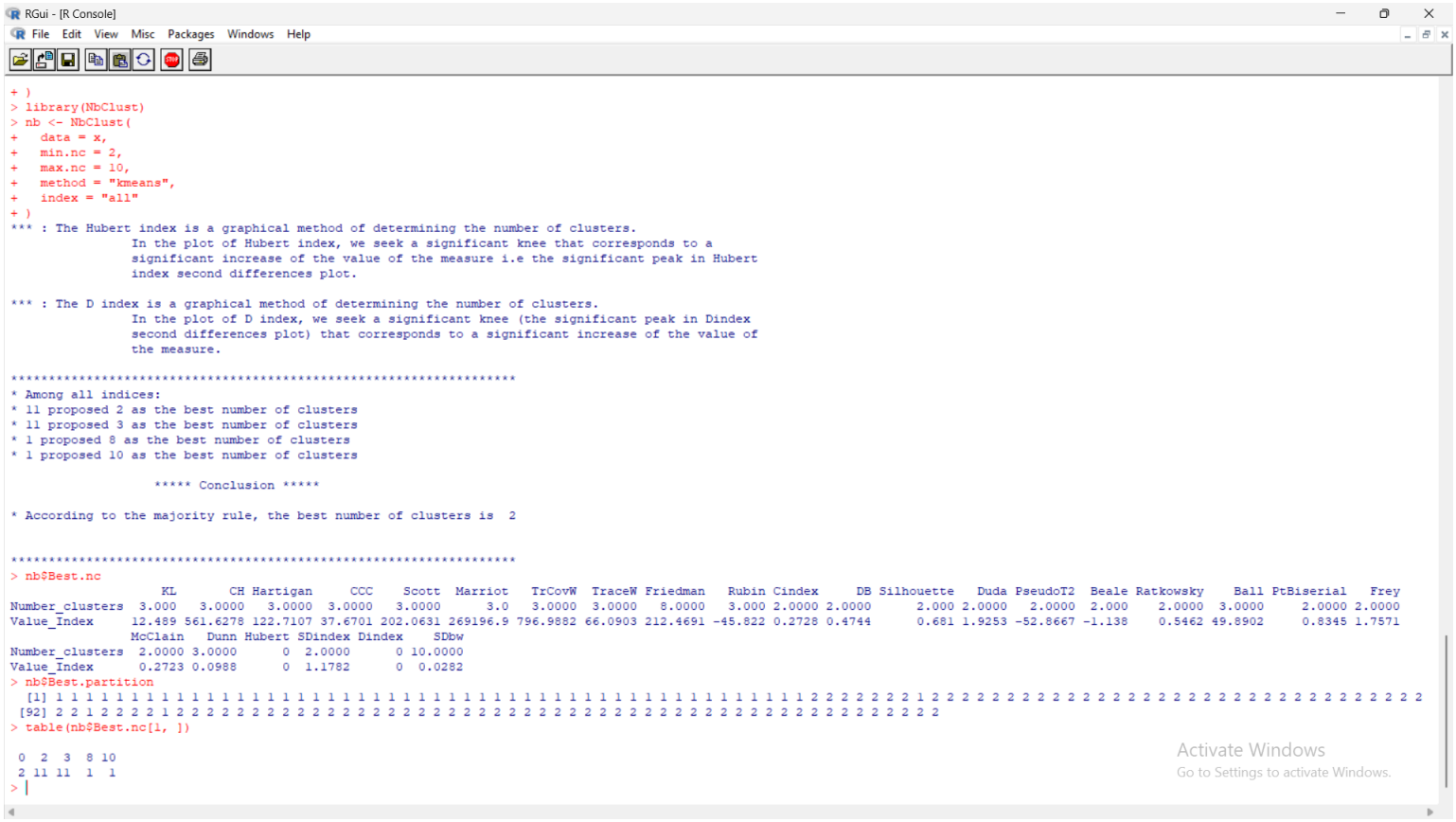
b) From your plot, identify the k value at the 'elbow' (bend) of the curve. Does this agree with the known number of iris species?

Notice that the WSS **always decreases** as k increases. That's expected.

If you increase the number of clusters, each cluster generally contains fewer observations, so the observations can be closer to their respective centroids.

The important thing is **not simply choosing the lowest WSS**. If we did that, we'd always choose k = 10 (or an even larger number). Instead, we look for the **elbow** — where adding more clusters gives progressively smaller improvements (it happened in cluster 3)

(c) Load the NbClust package and run NbClust() on the iris feature data (with the Species column removed) for min.nc = 2, max.nc = 10, method = "kmeans", index = "all". According to the majority rule, what is the best number of clusters?


RGui - [R Console]
File Edit View Misc Packages Windows Help
+)
> library(NbClust)
> nb <- NbClust(
+ data = x,
+ min.nc = 2,
+ max.nc = 10,
+ method = "kmeans",
+ index = "all"
+)
*** : The Hubert index is a graphical method of determining the number of clusters.
In the plot of Hubert index, we seek a significant knee that corresponds to a
significant increase of the value of the measure i.e the significant peak in Hubert
index second differences plot.
*** : The D index is a graphical method of determining the number of clusters.
In the plot of D index, we seek a significant knee (the significant peak in Dindex
second differences plot) that corresponds to a significant increase of the value of
the measure.
* Among all indices:
* 11 proposed 2 as the best number of clusters
* 11 proposed 3 as the best number of clusters
* 1 proposed 8 as the best number of clusters
* 1 proposed 10 as the best number of clusters
***** Conclusion *****
* According to the majority rule, the best number of clusters is 2
> nb$Best.nc
KL CH Hartigan CCC Scott Marriot TrCovW TraceW Friedman Rubin Cindex DB Silhouette Duda PseudoT2 Beale Ratkowsky Ball PtBiserial Frey
Number_clusters 3.000 3.0000 3.0000 3.0000 3.0000 3.0 3.0000 3.0000 8.0000 3.000 2.0000 2.0000 2.000 2.0000 2.0000 2.000 2.0000 3.0000 2.0000 2.0000
Value_Index 12.489 561.6278 122.7107 37.6701 202.0631 269196.9 796.9882 66.0903 212.4691 -45.822 0.2728 0.4744 0.681 1.9253 -52.8667 -1.138 0.5462 49.8902 0.8345 1.7571
McClain Dunn Hubert SDindex Dindex SDbw
Number_clusters 2.0000 3.0000 0 2.0000 0 10.0000
Value_Index 0.2723 0.0988 0 1.1782 0 0.0282
> nb$Best.partition
[1] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 1 2 2 2 2 2 2 2

(d) Briefly explain, in 2–3 sentences, why a single index (like the elbow plot) might disagree with the majority-vote result from NbClust, and why it can be useful to consult multiple indices.

A single method such as the elbow plot may recommend a different number of clusters because it evaluates cluster quality using one specific criterion, while NbClust combines the results of many different clustering indices. Consulting multiple indices is useful because different measures assess different aspects of cluster quality, providing a more reliable basis for choosing the number of clusters.

this is demonstrated by your results:

Elbow method → k = 3

NbClust majority rule → k = 2

(both of these methods are unsupervised clustering, because They do **not use the true Species labels)**

Part B — K-Means on the Wine Dataset

Question 6. Loading and exploring winedata.csv

(a) Read winedata.csv into a data frame called wine using read.csv()

```
> wine <- read.csv("winedata.csv")
> head(wine)
  Type Alcohol Malic  Ash Alcalinity Magnesium Phenols Flavanoids Nonflavanoids Proanthocyanins Color  Hue Dilution Proline
1    1   14.23  1.71 2.43       15.6       127    2.80       3.06          0.28            2.29  5.64 1.04     3.92    1065
2    1   13.20  1.78 2.14       11.2       100    2.65       2.76          0.26            1.28  4.38 1.05     3.40    1050
3    1   13.16  2.36 2.67       18.6       101    2.80       3.24          0.30            2.81  5.68 1.03     3.17    1185
4    1   14.37  1.95 2.50       16.8       113    3.85       3.49          0.24            2.18  7.80 0.86     3.45    1480
5    1   13.24  2.59 2.87       21.0       118    2.80       2.69          0.39            1.82  4.32 1.04     2.93     735
6    1   14.20  1.76 2.45       15.2       112    3.27       3.39          0.34            1.97  6.75 1.05     2.85    1450
>
```

(b) Use str() to inspect the structure of wine. How many observations and how many variables are there? Which variable represents the wine's true class (cultivar)?

```
> str(wine)
'data.frame':   178 obs. of  14 variables:
 $ Type           : int  1 1 1 1 1 1 1 1 1 1 ...
 $ Alcohol        : num  14.2 13.2 13.2 14.4 13.2 ...
 $ Malic          : num  1.71 1.78 2.36 1.95 2.59 1.76 1.87 2.15 1.64 1.35 ...
 $ Ash            : num  2.43 2.14 2.67 2.5 2.87 2.45 2.45 2.61 2.17 2.27 ...
 $ Alcalinity     : num  15.6 11.2 18.6 16.8 21 15.2 14.6 17.6 14 16 ...
 $ Magnesium      : int  127 100 101 113 118 112 96 121 97 98 ...
 $ Phenols        : num  2.8 2.65 2.8 3.85 2.8 3.27 2.5 2.6 2.8 2.98 ...
 $ Flavanoids     : num  3.06 2.76 3.24 3.49 2.69 3.39 2.52 2.51 2.98 3.15 ...
 $ Nonflavanoids  : num  0.28 0.26 0.3 0.24 0.39 0.34 0.3 0.31 0.29 0.22 ...
 $ Proanthocyanins: num  2.29 1.28 2.81 2.18 1.82 1.97 1.98 1.25 1.98 1.85 ...
 $ Color          : num  5.64 4.38 5.68 7.8 4.32 6.75 5.25 5.05 5.2 7.22 ...
 $ Hue            : num  1.04 1.05 1.03 0.86 1.04 1.05 1.02 1.06 1.08 1.01 ...
 $ Dilution       : num  3.92 3.4 3.17 3.45 2.93 2.85 3.58 3.58 2.85 3.55 ...
 $ Proline        : int  1065 1050 1185 1480 735 1450 1290 1295 1045 1045 ...
>
```

178 observations → 178 wine samples

14 variables → 14 columns

13 chemical variables

1 true class variable(cultivar): Type

c) Using summary() or otherwise, comment on whether the 13 chemical variables are all measured on similar scales.

```
> summary(wine)
      Type          Alcohol         Malic            Ash          Alcalinity      Magnesium        Phenols        Flavanoids    Nonflavanoids   Proanthocyanins      Color      
 Min.   :1.000   Min.   :11.03   Min.   :0.740   Min.   :1.360   Min.   :10.60   Min.   : 70.00   Min.   :0.980   Min.   :0.340   Min.   :0.1300   Min.   :0.410   Min.   : 1.280  
 1st Qu.:1.000   1st Qu.:12.36   1st Qu.:1.603   1st Qu.:2.210   1st Qu.:17.20   1st Qu.: 88.00   1st Qu.:1.742   1st Qu.:1.205   1st Qu.:0.2700   1st Qu.:1.250   1st Qu.: 3.220  
 Median :2.000   Median :13.05   Median :1.865   Median :2.360   Median :19.50   Median : 98.00   Median :2.355   Median :2.135   Median :0.3400   Median :1.555   Median : 4.690  
 Mean   :1.938   Mean   :13.00   Mean   :2.336   Mean   :2.367   Mean   :19.49   Mean   : 99.74   Mean   :2.295   Mean   :2.029   Mean   :0.3619   Mean   :1.591   Mean   : 5.058  
 3rd Qu.:3.000   3rd Qu.:13.68   3rd Qu.:3.083   3rd Qu.:2.558   3rd Qu.:21.50   3rd Qu.:107.00   3rd Qu.:2.800   3rd Qu.:2.875   3rd Qu.:0.4375   3rd Qu.:1.950   3rd Qu.: 6.200  
 Max.   :3.000   Max.   :14.83   Max.   :5.800   Max.   :3.230   Max.   :30.00   Max.   :162.00   Max.   :3.880   Max.   :5.080   Max.   :0.6600   Max.   :3.580   Max.   :13.000  
      Hue            Dilution        Proline      
 Min.   :0.4800   Min.   :1.270   Min.   : 278.0  
 1st Qu.:0.7825   1st Qu.:1.938   1st Qu.: 500.5  
 Median :0.9650   Median :2.780   Median : 673.5  
 Mean   :0.9574   Mean   :2.612   Mean   : 746.9  
 3rd Qu.:1.1200   3rd Qu.:3.170   3rd Qu.: 985.0  
 Max.   :1.7100   Max.   :4.000   Max.   :1680.0  
> 
```

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The scales are **very different**.

For example:

Alcohol → 11.03 to 14.83

Magnesium → 70 to 162

Proline → 278 to 1680

while some variables are only around 0–5.

Why does this matter for K-means?

K-means uses **distance** to determine which observations belong to the same cluster.

Because Proline has values as high as **1680**, it can have a much greater influence on the distance than something like Hue, which ranges only from **0.48 to 1.71**.

Therefore, we need to **scale the variables** before clustering.

Question 7. Scaling the data

(a) Create a scaled version of the chemical variables (excluding the Type column) called data.train, using the scale() function.

```
> data.train <- scale(wine[, -1])
> head(data.train)
       Alcohol       Malic         Ash Alcalinity  Magnesium    Phenols Flavanoids Nonflavanoids Proanthocyanins      Color        Hue  Dilution     Proline
[1,] 1.5143408 -0.56066822  0.2313998 -1.1663032 1.90852151 0.8067217  1.0319081    -0.6577078       1.2214385  0.2510088  0.3611585 1.8427215  1.01015939
[2,] 0.2455968 -0.49800856 -0.8256672 -2.4838405 0.01809398 0.5670481  0.7315653    -0.8184106      -0.5431887 -0.2924962  0.4049085 1.1103172  0.96252635
[3,] 0.1963252  0.02117152  1.1062139 -0.2679823 0.08810981 0.8067217  1.2121137    -0.4970050       2.1299594  0.2682629  0.3174085 0.7863692  1.39122370
[4,] 1.6867914 -0.34583508  0.4865539 -0.8069748 0.92829983 2.4844372  1.4623994    -0.9791134       1.0292513  1.1827317 -0.4263410 1.1807407  2.32800680
[5,] 0.2948684  0.22705328  1.8352256  0.4506745 1.27837900 0.8067217  0.6614853     0.2261576       0.4002753 -0.3183774  0.3611585 0.4483365 -0.03776747
[6,] 1.4773871 -0.51591132  0.3043010 -1.2860793 0.85828399 1.5576991  1.3622851    -0.1755994       0.6623487  0.7298108  0.4049085 0.3356589  2.23274072
> |
```

(b) Explain why scaling is necessary before applying k-means to this dataset, referring to how k means computes distances.

K means uses distance to decide which observations are closer to each other.

For example, it calculates distances using the chemical measurements: Alcohol, malic

A simplified Euclidean distance is:

$d=[(x_1-y_1)^2+(x_2-y_2)^2+\cdots+(x_{13}-y_{13})^2]^{1/2}$

Example:

Wine A = (1, 2, 3)

Centroid = (2, 4, 5)

$d^2=(1-2)^2+(2-4)^2+(3-5)^2$

d=3

So the **squared distance = 9**

The actual Euclidean distance(d) would be: $9^{1/2}=3$

Example:

So the problem is :

Alcohol → 11.03 – 14.83

Malic → 0.74 – 5.80

Magnesium → 70 – 162

Proline → 278 – 1680

Because Proline has much larger numerical values, it can contribute **much more to the distance calculation** than variables such as Alcohol or Hue

Example:

Suppose two wines differ by:

Alcohol: 1

Proline: 300

In the squared-distance calculation:

$1^2=1$

but:

$300^2=90{,}000$

So the Proline difference can have a dramatically larger effect on the distance.

That's the main reason we scale the data.

(c) Confirm that each scaled variable now has a mean of (approximately) 0 using summary(data.train) or colMeans()

```
> colMeans(data.train)
      Alcohol         Malic           Ash     Alcalinity     Magnesium       Phenols     Flavanoids Nonflavanoids Proanthocyanins         Color           Hue
-8.594093e-16  -6.734236e-17   8.046486e-16  -7.684922e-17  -4.095117e-17  -1.391677e-17   6.950589e-17  -1.041614e-16  -1.223272e-16   3.676054e-17   2.100087e-16
     Dilution       Proline
 3.009648e-16  -1.037131e-16
>
```

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The column means of the scaled variables are all approximately 0. The very small values, such as −8.59×10−16 and 3.01×10−16, are effectively zero and result from floating-point numerical precision. Therefore, the scale() function has successfully standardized the 13 chemical variables

Question 8. Choosing k and fitting the model

(a) Run NbClust on data.train (min.nc = 2, max.nc = 15, method = "kmeans") and produce a bar chart of the number of criteria proposing each value of k

```
> library(NbClust)
> nb.wine <- NbClust(
+   data = data.train,
+   min.nc = 2,
+   max.nc = 15,
+   method = "kmeans",
+   index = "all"
+ )
*** : The Hubert index is a graphical method of determining the number of clusters.
                In the plot of Hubert index, we seek a significant knee that corresponds to a
                significant increase of the value of the measure i.e the significant peak in Hubert
                index second differences plot.

*** : The D index is a graphical method of determining the number of clusters.
                In the plot of D index, we seek a significant knee (the significant peak in Dindex
                second differences plot) that corresponds to a significant increase of the value of
                the measure.

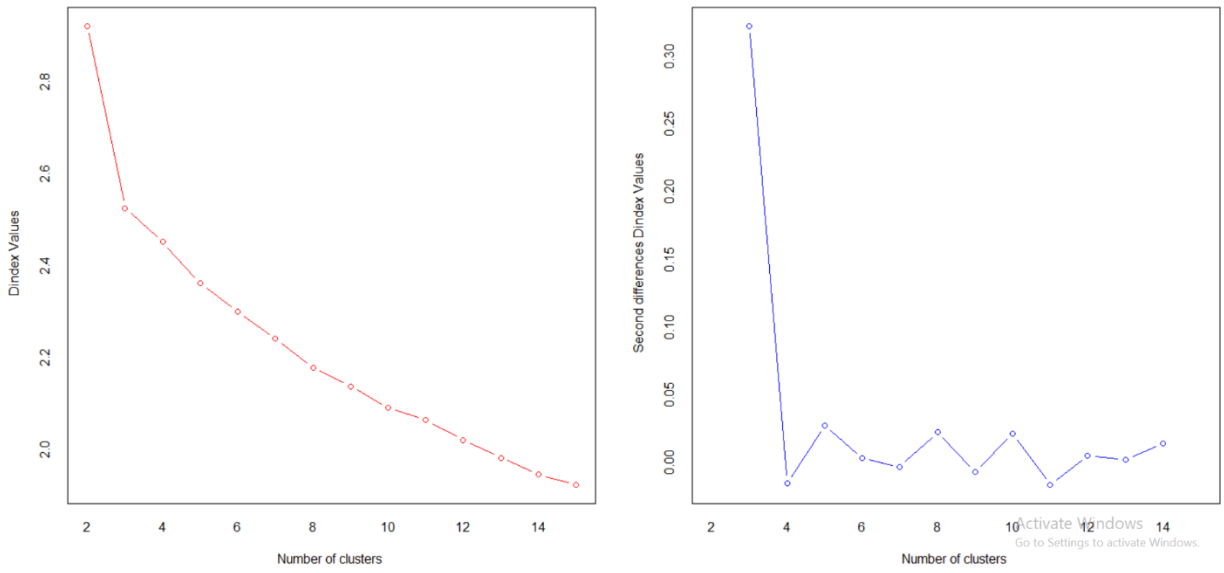
*******************************************************************
* Among all indices:
* 2 proposed 2 as the best number of clusters
* 19 proposed 3 as the best number of clusters
* 1 proposed 14 as the best number of clusters
* 1 proposed 15 as the best number of clusters

                   ***** Conclusion *****

* According to the majority rule, the best number of clusters is  3


*******************************************************************
>
```

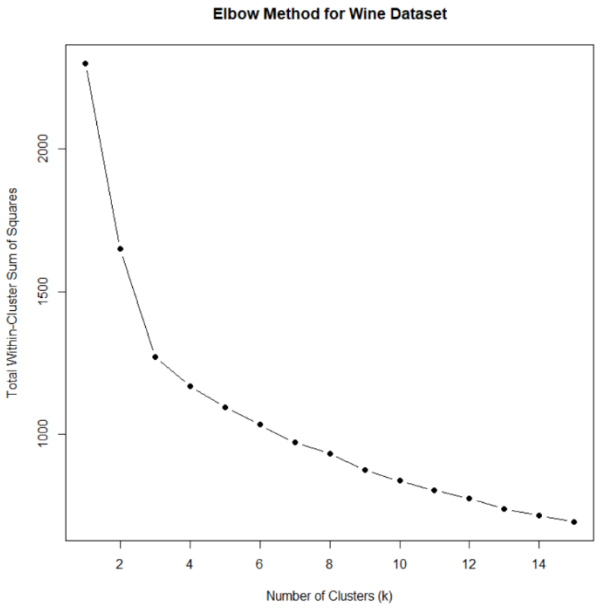
NbClust was applied to the scaled wine dataset using k-means clustering for values of k from 2 to 15. Two criteria proposed k = 2, nineteen criteria proposed k = 3, one criterion proposed k = 14, and one criterion proposed k = 15. Therefore, according to the majority rule, k = 3 was selected as the best number of clusters.



(b) Separately, write a for loop that computes the within-cluster sum of squares for k = 1 to 15 and plot the resulting scree plot. Does the elbow agree with NbClust's recommendation?

```
> wss.wine <- numeric(15)
> set.seed(123)
>
> for (k in 1:15) {
+   km <- kmeans(data.train, centers = k, nstart = 20)
+   wss.wine[k] <- km$tot.withinss
+ }
> wss.wine
 [1] 2301.0000 1649.4400 1270.7491 1168.6143 1095.1529 1033.9668  970.7864  931.1976  874.8905  837.1795  803.5661  775.0478  738.9021  715.8005  692.7357
> plot(
+   1:15,
+   wss.wine,
+   type = "b",
+   pch = 19,
+   xlab = "Number of Clusters (k)",
+   ylab = "Total Within-Cluster Sum of Squares",
+   main = "Elbow Method for Wine Dataset"
+ )
```

The scree plot shows that the total within-cluster sum of squares decreases as the number of clusters increases. The largest reductions occur between k = 1 and k = 3, after which the decrease becomes considerably smaller. Therefore, the elbow appears at approximately k = 3. This agrees with the NbClust majority-rule recommendation, which also selected k = 3.



(c) Set a seed and fit a final k-means model, fit.km, to data.train using the number of clusters you decided on in parts (a)-(b).

```
> set.seed(123)
> fit.km <- kmeans(
+   data.train,
+   centers = 3,
+   nstart = 20
+ )
> fit.km
K-means clustering with 3 clusters of sizes 51, 62, 65

Cluster means:
     Alcohol      Malic        Ash Alcalinity   Magnesium     Phenols
1  0.1644436  0.8690954  0.1863726  0.5228924 -0.07526047 -0.97657548
2  0.8328826 -0.3029551  0.3636801 -0.6084749  0.57596208  0.88274724
3 -0.9234669 -0.3929331 -0.4931257  0.1701220 -0.49032869 -0.07576891
   Flavanoids Nonflavanoids Proanthocyanins      Color        Hue   Dilution
1 -1.21182921    0.72402116     -0.77751312  0.9388902 -1.1615122 -1.2887761
2  0.97506900   -0.56050853      0.57865427  0.1705823  0.4726504  0.7770551
3  0.02075402   -0.03343924      0.05810161 -0.8993770  0.4605046  0.2700025
     Proline
1 -0.4059428
2  1.1220202
3 -0.7517257

Clustering vector:
  [1] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
 [38] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 3 3 1 3 3 3 3 3 3 3 3 3 3 3 3 2
 [75] 3 3 3 3 3 3 3 3 1 3 3 3 3 3 3 3 3 3 3 3 2 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
[112] 3 3 3 3 3 3 3 1 3 3 2 3 3 3 3 3 3 3 3 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
[149] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1

Within cluster sum of squares by cluster:
[1] 326.3537 385.6983 558.6971
 (between_SS / total_SS =  44.8 %)

Available components:

[1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss"
[6] "betweenss"    "size"         "iter"         "ifault"
> |
```

Based on both the NbClust majority rule and the elbow method, k = 3 was selected. A final K-means model was fitted to the scaled wine dataset using three clusters, a random seed of 123, and nstart = 20. The resulting clusters contained 51, 62, and 65 observations, respectively. The model achieved a between-cluster sum of squares to total sum of squares ratio of 44.8%.

Question 9. Interpreting and evaluating the clusters

a) Print fit.km$centers. Choose one cluster and describe, in a sentence or two, what kind of wine it represents (e.g. relatively high/low in particular chemicals).

```
> fit.km$centers
     Alcohol      Malic        Ash Alcalinity   Magnesium     Phenols  Flavanoids Nonflavanoids Proanthocyanins      Color        Hue   Dilution    Proline
1  0.1644436  0.8690954  0.1863726  0.5228924 -0.07526047 -0.97657548 -1.21182921    0.72402116     -0.77751312  0.9388902 -1.1615122 -1.2887761 -0.4059428
2  0.8328826 -0.3029551  0.3636801 -0.6084749  0.57596208  0.88274724  0.97506900   -0.56050853      0.57865427  0.1705823  0.4726504  0.7770551  1.1220202
3 -0.9234669 -0.3929331 -0.4931257  0.1701220 -0.49032869 -0.07576891  0.02075402   -0.03343924      0.05810161 -0.8993770  0.4605046  0.2700025 -0.7517257
> |
```

Activate Windows
Go to Settings to activate Windows.

Cluster 2 represents wines with relatively high Alcohol, Magnesium, Phenols, Flavanoids, Proanthocyanins, and Proline levels compared with the overall dataset average. It also has relatively low Malic Acid, Alcalinity, and Nonflavanoids levels.

b) Build a confusion matrix comparing wine$Type to fit.km$cluster using table(). How many wines are misclassified overall?

```
> table(wine$Type, fit.km$cluster)

    1  2  3
  1  0 59  0
  2  3  3 65
  3 48  0  0
> |
```

The confusion matrix shows that 59 Type 1 wines were assigned to Cluster 2, 65 Type 2 wines were assigned to Cluster 3, and 48 Type 3 wines were assigned to Cluster 1. Since cluster labels are arbitrary, the clusters were matched to the dominant wine types. Thus, 172 out of 178 wines were correctly clustered, resulting in 6 misclassified wines overall.

c) Load the flexclust package and compute the Adjusted Rand Index (ARI) for your confusion matrix using randIndex(). What does an ARI close to 1 indicate, and what would an ARI of 0 mean?

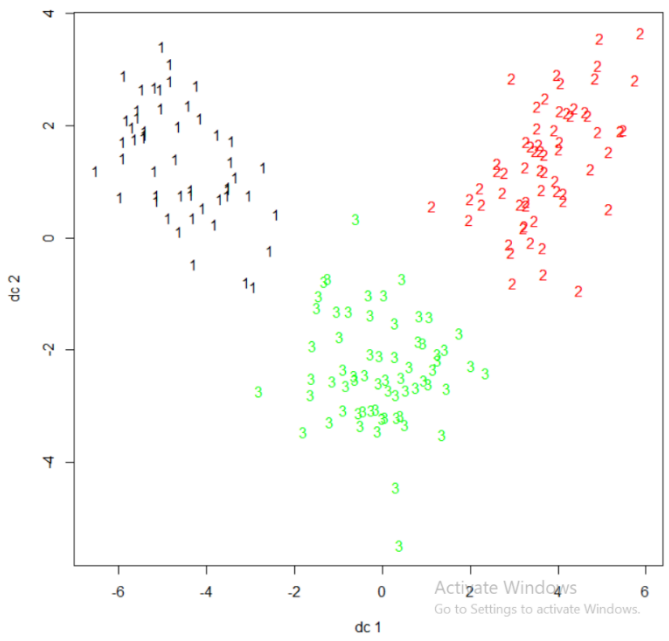
```
> library(flexclust)
> randIndex(table(wine$Type, fit.km$cluster))
      ARI
0.897495
>
```

The Adjusted Rand Index (ARI) for the K-means clustering is 0.897495. Since this value is close to 1, it indicates a strong agreement between the clusters produced by K-means and the known wine classes. An ARI of 0 would indicate that the agreement between the clustering and the true class labels is approximately equivalent to what would be expected by random chance.

Question 10 Visualisation

(a) Using the fpc package, produce a discriminant projection plot of the wine clusters with plotcluster().

```
> library(fpc)
> plotcluster(
+   data.train,
+   fit.km$cluster
+ )
>
```



b) In 2–3 sentences, comment on how well-separated the three clusters appear to be in this projection.

The discriminant projection shows that the three wine clusters are generally well separated, with most observations forming distinct groups. However, there is some overlap between the clusters, particularly for a small number of observations, which is consistent with the six misclassified wines and the ARI value of 0.8975.

Part C —Short Answer / Theory

Question 11. Conceptual questions

(a) List, in order, the four main steps of the k-means algorithm.
The four main steps of the K-means algorithm are:

1. **Initialize** k cluster centroids.
2. **Assign** each observation to the nearest centroid based on distance.
3. **Update** each centroid by calculating the mean of the observations assigned to that cluster.
4. **Repeat** the assignment and update steps until the clusters converge or the stopping condition is reached.

(b) K-means is described as a 'partitioning' method. Name one clustering method that is NOT a partitioning method, and briefly state one key difference between it and k-means.

Hierarchical clustering is a clustering method that is not a partitioning method. Unlike K-means, which directly divides the data into a predefined number of clusters by assigning observations to the nearest centroid, hierarchical clustering builds a hierarchy of nested clusters by repeatedly merging or splitting clusters. The resulting hierarchy can be visualised using a dendrogram.

(c) Give one advantage and one limitation of k-means clustering compared to hierarchical clustering.

Advantage: K-means is generally faster and more computationally efficient than hierarchical clustering, especially when working with large datasets.

Limitation: K-means requires the number of clusters (k) to be specified in advance, whereas hierarchical clustering can produce a dendrogram that allows different numbers of clusters to be explored without initially specifying k.

(d) Why is k-means considered an unsupervised learning technique, even though the datasets you used in this lab (iris and wine) actually have known class labels?

K-means is considered an **unsupervised learning** technique because it creates clusters using only the input features and does not use the known class labels during the clustering process. Although the Iris and Wine datasets contain true labels such as Species and Type, these labels were only used after clustering to evaluate how well the resulting clusters matched the known classes. Therefore, K-means itself remains unsupervised.